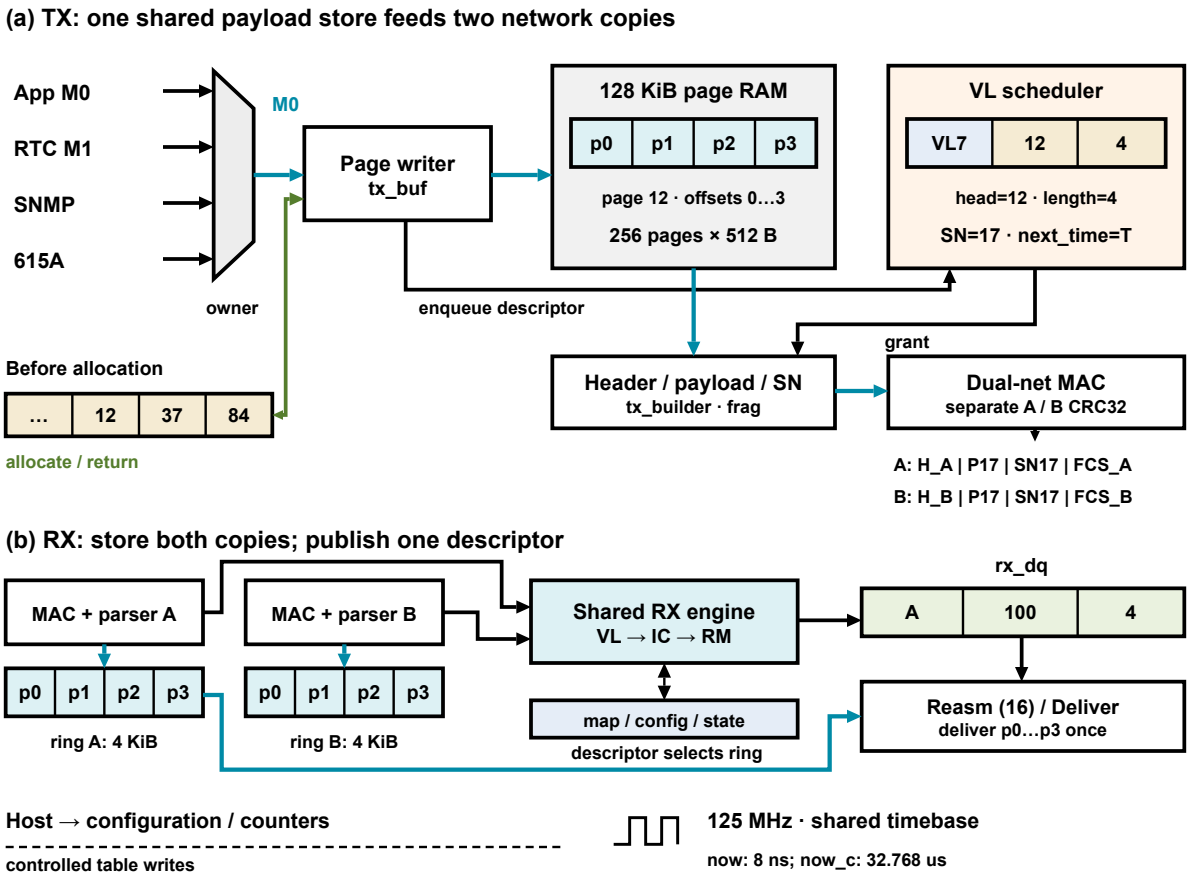
(a) TX: one shared payload store feeds two network copies
App M0
RTC M1
SNMP
615A
M0
owner
Page writer
tx_buf
128 KiB page RAM
p0
p1
p2
p3
page 12 · offsets 0…3
256 pages × 512 B
VL scheduler
VL7
12
4
head=12 · length=4
SN=17 · next_time=T
enqueue descriptor
grant
Before allocation
…
12
37
84
allocate / return
Header / payload / SN
tx_builder · frag
Dual-net MAC
separate A / B CRC32
A: H_A | P17 | SN17 | FCS_A
B: H_B | P17 | SN17 | FCS_B
(b) RX: store both copies; publish one descriptor
MAC + parser A
MAC + parser B
Shared RX engine
VL → IC → RM
rx_dq
A
100
4
p0
p1
p2
p3
ring A: 4 KiB
ring B: 4 KiB
map / config / state
descriptor selects ring
Reasm (16) / Deliver
deliver p0…p3 once
Host → configuration / counters
controlled table writes
125 MHz · shared timebase
now: 8 ns; now_c: 32.768 us